

INFO531/ISTA431: Data Warehousing and Analytics in the Cloud

Topic: Stock Evaluation

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In the entire project, we will use stock prices from the following excel file:

https://docs.google.com/spreadsheets/d/17141X9cmwzhjcmQ-tz5UM3t5D4ZBN3x7/edit?usp=share_link&ouid=113826847320039112892&rtpof=true&sd=true

Create a SQL Server database in azure. Here are the steps that we have followed to create a SQL Server database.

Create SQL Database ...

Microsoft

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'projectdb0' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

- Connectivity method * ⓘ
- ☐ No access
 - ☒ Public endpoint
 - ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

☒ No ☐ Yes

Add current client IP address *

☐ No ☒ Yes

Connection policy

Configure how clients communicate with your SQL database server. [Learn more](#)

- Connection policy ⓘ
- ☒ Default - Uses Redirect policy for all client connections originating inside of Azure and Proxy for all client connections originating outside Azure
 - ☐ Proxy - All connections are proxied via the Azure SQL Database gateways
 - ☐ Redirect - Clients establish connections directly to the node hosting the database

Encrypted connections

This server supports encrypted connections using Transport Layer Security (TLS). For information on TLS version and certificates, refer to connecting with TLS/SSL. [Learn more](#)

Minimum TLS version ⓘ



Cost summary

General Purpose (GP_Gen5_2)	
Cost per vCore (in USD)	184.09
vCores selected	x 2
Cost per GB (in USD)	0.12
Max storage selected (in GB)	x 41.6
ESTIMATED COST / MONTH	372.97 USD

[Review + create](#)

[< Previous](#)

[Next : Security >](#)

Create SQL Database ...

Microsoft

Basics Networking Security Additional settings Tags Review + create

Microsoft Defender for SQL

Protect your data using Microsoft Defender for SQL, a unified security package including vulnerability assessment and advanced threat protection for your server. [Learn more](#)

Get started with a 30 day free trial period, and then 15 USD/server/month.

Enable Microsoft Defender for SQL * ⓘ ☐ Start free trial
☒ Not now

Ledger

Ledger cryptographically verifies the integrity of your data and detects any tampering that might have occurred. [Learn more](#)

Ledger **Not configured**
[Configure ledger](#)

Identity

Use system-assigned and user-assigned managed identities to enable central access management between this database and other Azure resources. [Learn more](#)

Identity **Not enabled**
[Configure Identities](#)

Transparent data encryption

Transparent data encryption (TDE) encrypts your databases, backups, and logs at rest without any changes to your application. [Learn more](#)

Transparent data encryption ⓘ **Service-managed key selected**
[Configure transparent data encryption](#)



Cost summary

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[Home](#) >

Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data * **None** Backup Sample

Database collation

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. [Learn more](#)

Collation * ⓘ [Find a collation](#)

Maintenance window

Select a preferred maintenance window from the drop down. Please note, during a maintenance event, Azure SQL Database are fully available and accessible but some of the maintenance updates require a failover as Azure takes SQL DB instances offline for a short time to apply the maintenance updates. If the database is part of elastic pool, the maintenance configuration of elastic pool will be applied. [Learn more](#)

Maintenance window ▼



Cost summary

General Purpose (GP_Gen5_2)	
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vCores selected	x 2
Cost per GB (in USD)	0.12
Max storage selected (in GB)	x 41.6
ESTIMATED COST / MONTH	372.97 USD

[Home](#) >

Create SQL Database ...

Microsoft

Basics Networking Security Additional settings **Tags** Review + create

Tags are name/value pairs that enable you to categorize and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more](#) 

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name ⓘ	Value ⓘ	Resource
	:	2 selected 



Cost summary

General Purpose (GP_Gen5_2)	
Cost per vCore (in USD)	184.09
vCores selected	x 2
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ESTIMATED COST / MONTH	372.97 USD

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[Next : Review + create >](#)

Microsoft Azure

Home >

Create SQL Database

Microsoft

BasicsNetworkingSecurityAdditional settingsTagsReview + create

Product details

SQL database
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost per month
372.97 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription

Resource group

Region

Database name

Server

Authentication method

Server admin login

Azure AD Admin

Compute + storage

Backup storage redundancy

Azure for Students

azuresql

East US

projectdb

(new) projectdb0

SQL and Azure Active Directory authentication

ssaa

ripashah@arizona.edu

General Purpose: Standard-series (Gen5), 2 vCores, 32 GB storage, zone redundant disabled

Geo-redundant backup storage

Networking

Allow Azure services and resources to access this server

Add current client IP address 76.50.32.31

Private endpoint

Minimum TLS version

Connection Policy

No

Yes

None

1.2

Default

Create

< Previous

[Download a template for automation](#)



Cost summary

General Purpose (GP_Gen5_2)

Cost per vCore (in USD)

184.09

vCores selected

x 2

Cost per GB (in USD)

0.12

Max storage selected (in GB)

x 41.6

ESTIMATED COST / MONTH

372.97 USD

Microsoft Azure

Search resources, services, and docs (G+)

Home >

Microsoft.SQLDatabase.newDatabaseNewServer_b7dcbc0764fa4456b3ef9 | Overview

Deployment

Search

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name: Microsoft.SQLDatabase.newDatabaseNewServe...

Subscription: Azure for Students

Resource group: azuresql

Start time: 11/21/2022, 9:37:41 PM

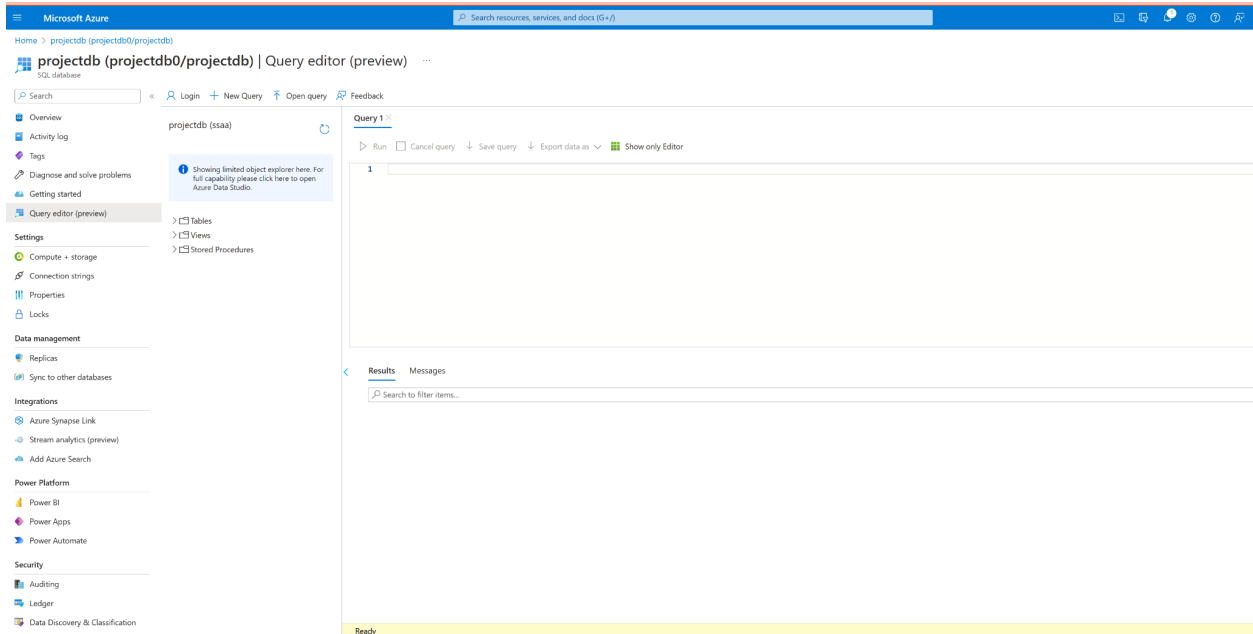
Correlation ID: 7e662366-9255-4fcf-8a48-d942f3a8d37b

Deployment details

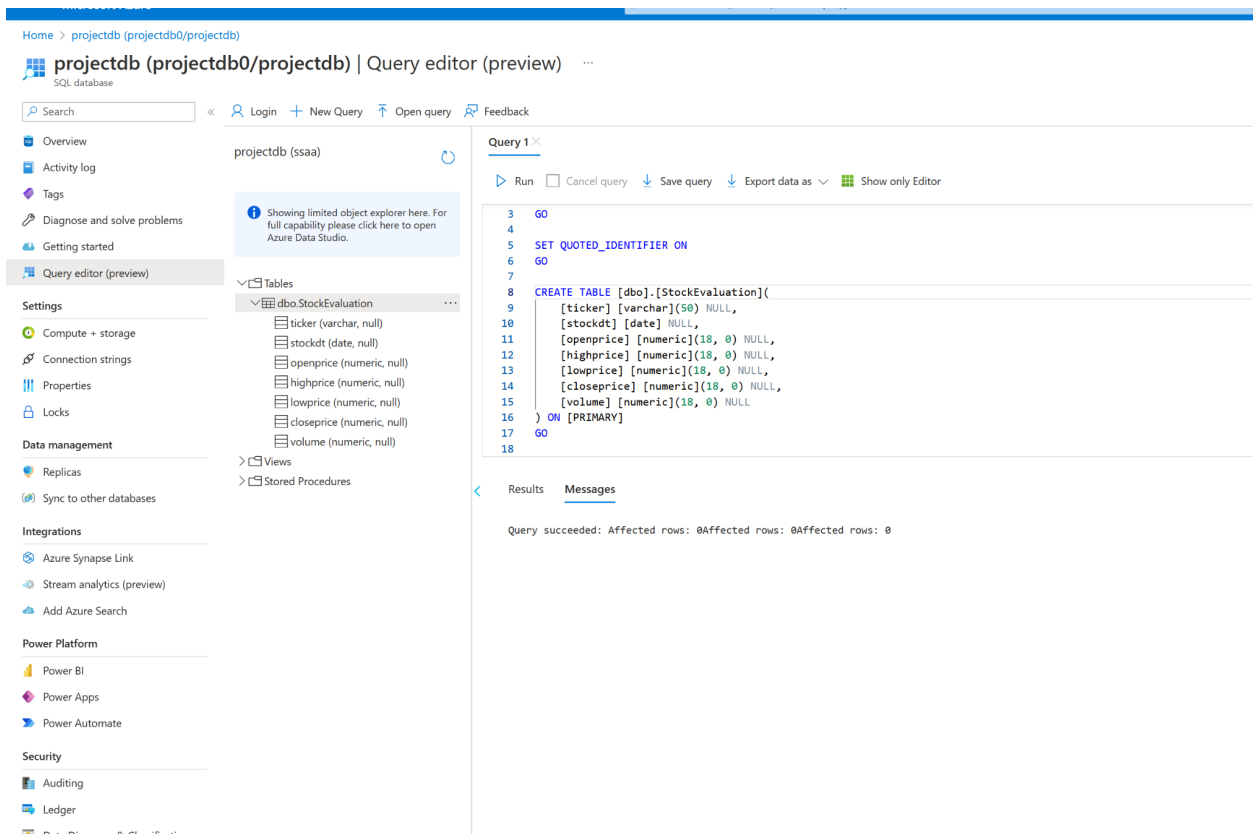
Resource	Type	Status	Operation details
projectdb0	Microsoft.Sql/servers	Accepted	Operation details

Give feedback

Tell us about your experience with deployment



Once the SQL Server database is created. You should create a staging table first.



```

/***** Object: Table [dbo].[StockEvaluation] Script Date: 12/4/2022
6:53:51 PM *****/

```

```
SET ANSI_NULLS ON
GO

SET QUOTED_IDENTIFIER ON
GO

CREATE TABLE [dbo].[StockEvaluation](
    [ticker] [varchar](50) NULL,
    [stockdt] [date] NULL,
    [openprice] [numeric](18, 0) NULL,
    [highprice] [numeric](18, 0) NULL,
    [lowprice] [numeric](18, 0) NULL,
    [closeprice] [numeric](18, 0) NULL,
    [volume] [numeric](18, 0) NULL,
    [id] [varchar](100) NULL
) ON [PRIMARY]
GO
```

Second, I have created SQL Server target table in my local instance by connecting to projectdb0.database.windows.net instance and entered my username and password using SQL Server authentication method.

```

/***** Object: Table [dbo].[StockEvaluation_Target]    Script Date: 12/4/2022
SET ANSI_NULLS ON
GO

```

```

SET QUOTED_IDENTIFIER ON
GO

```

```

CREATE TABLE [dbo].[StockEvaluation_Target](
    [ticker] [varchar](50) NULL,
    [stockdt] [date] NULL,
    [openprice] [numeric](18, 0) NULL,
    [highprice] [numeric](18, 0) NULL,
    [lowprice] [numeric](18, 0) NULL,
    [closeprice] [numeric](18, 0) NULL,
    [volume] [numeric](18, 0) NULL,
    [SourceSystemNM] [nvarchar](50) NOT NULL,
    [CreateAgentId] [nvarchar](20) NOT NULL,
    [CreateDtm] [datetime2](7) NOT NULL,
    [ChangeAgentId] [nvarchar](20) NOT NULL,
    [ChangeDtm] [datetime2](7) NOT NULL,
    [id] [varchar](100) NULL
) ON [PRIMARY]
GO

```

**** Object: Table [dbo].[StockEvaluation_Target] Script Date: 12/4/2022 6:52:16 PM ****/

```

SET ANSI_NULLS ON
GO

```

```

SET QUOTED_IDENTIFIER ON
GO

```

```

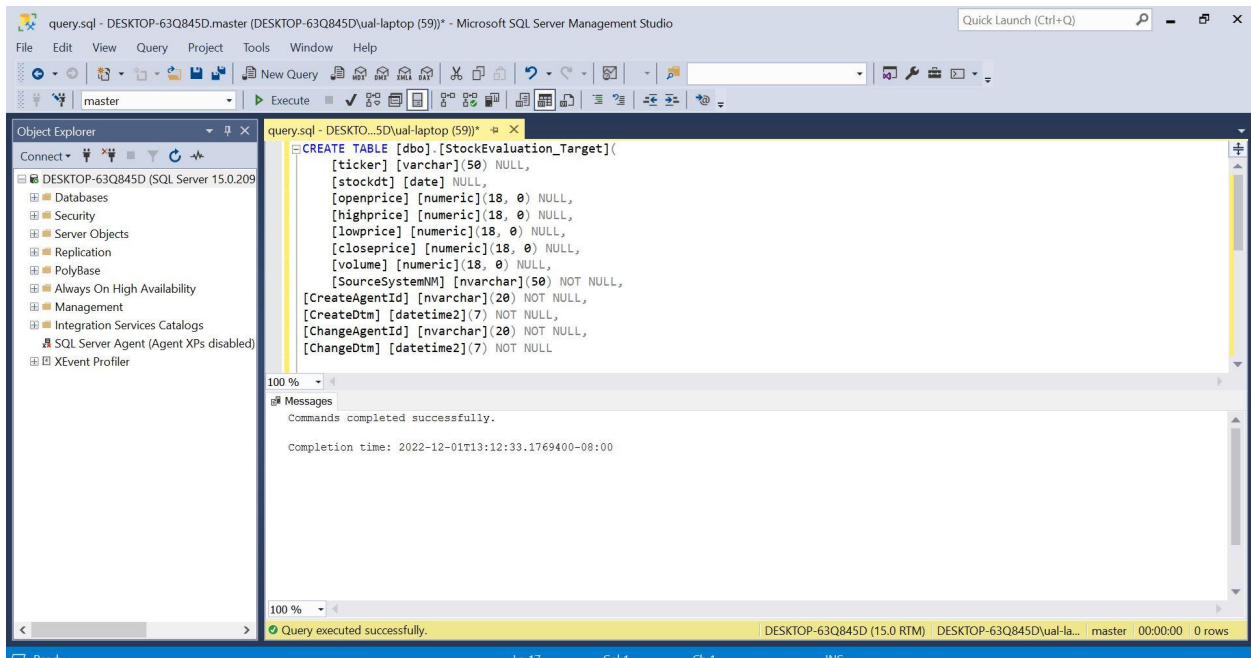
CREATE TABLE [dbo].[StockEvaluation_Target](

```

```

[ticker] [varchar](50) NULL,
[stockdt] [date] NULL,
[openprice] [numeric](18, 0) NULL,
[highprice] [numeric](18, 0) NULL,
[lowprice] [numeric](18, 0) NULL,
[closeprice] [numeric](18, 0) NULL,
[volume] [numeric](18, 0) NULL,
[SourceSystemNM] [nvarchar](50) NOT NULL,
[CreateAgentId] [nvarchar](20) NOT NULL,
[CreateDtm] [datetime2](7) NOT NULL,
[ChangeAgentId] [nvarchar](20) NOT NULL,
[ChangeDtm] [datetime2](7) NOT NULL,
[id] [varchar](100) NULL
) ON [PRIMARY]
GO

```



Create a container for blob storage

Microsoft Azure

Search resources, services, and docs (G+I)

Home > projectstocks

projectstocks | Containers

Storage account

Search

+ Container Change access level Restore containers Refresh Delete

Search containers by prefix

Show deleted containers

Name	Last modified	Public access level
☐ Slugs	12/2/2022, 9:15:30 AM	Private
☐ azureblob	12/2/2022, 9:18:24 AM	Blob

New container

Name *

azureblob1

Public access level ⓘ

Blob (anonymous read access for blobs only)

⚠ Blobs within the container can be read by anonymous request, but container data is not available. Anonymous clients cannot enumerate the blobs within the container.

Advanced

Create Discard

Upload a blob file in the container.

Microsoft Azure

Search resources, services, and docs (G+I)

Home > projectstocks > Containers >

azureblob

Container

Search

Upload Change access level Refresh Delete Change tier Acquire lease Break lease View snapshots Create snapshot

Authentication method: Access key (Switch to Azure AD User Account)

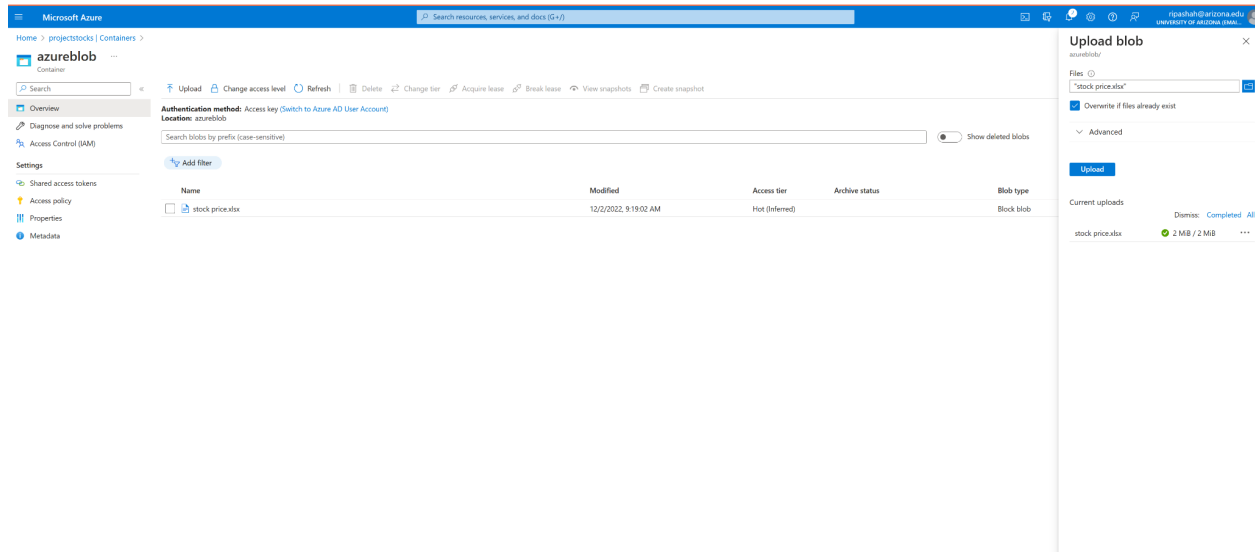
Location: azureblob

Search blobs by prefix (case sensitive)

Show deleted blobs

Add filter

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
☐ stock price.xlsx	12/2/2022, 9:19:02 AM	Hot (Inferred)		Block blob	1.54 MiB	Available



For further user interface, create an azure data factory

Microsoft Azure

Welcome to Azure Data Factory

☒ Create a new data factory ☐ Select an existing data factory

Create a new data factory

Name your factory, then pick a location and subscription to deploy in. For advanced options, please use [Azure Portal](#)

Name

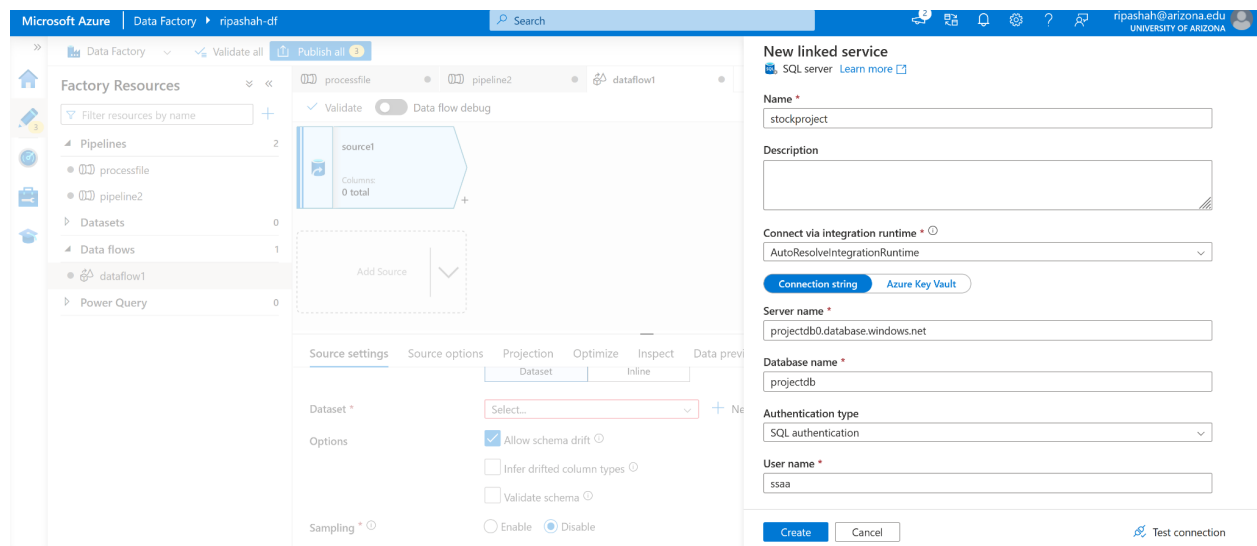
Location

Subscription

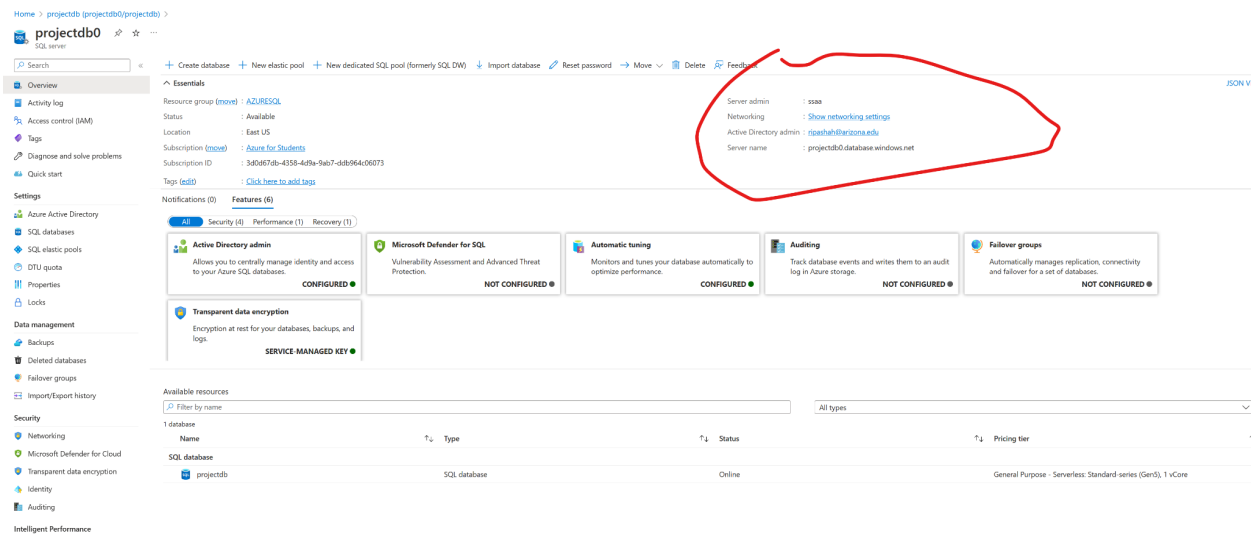
Create

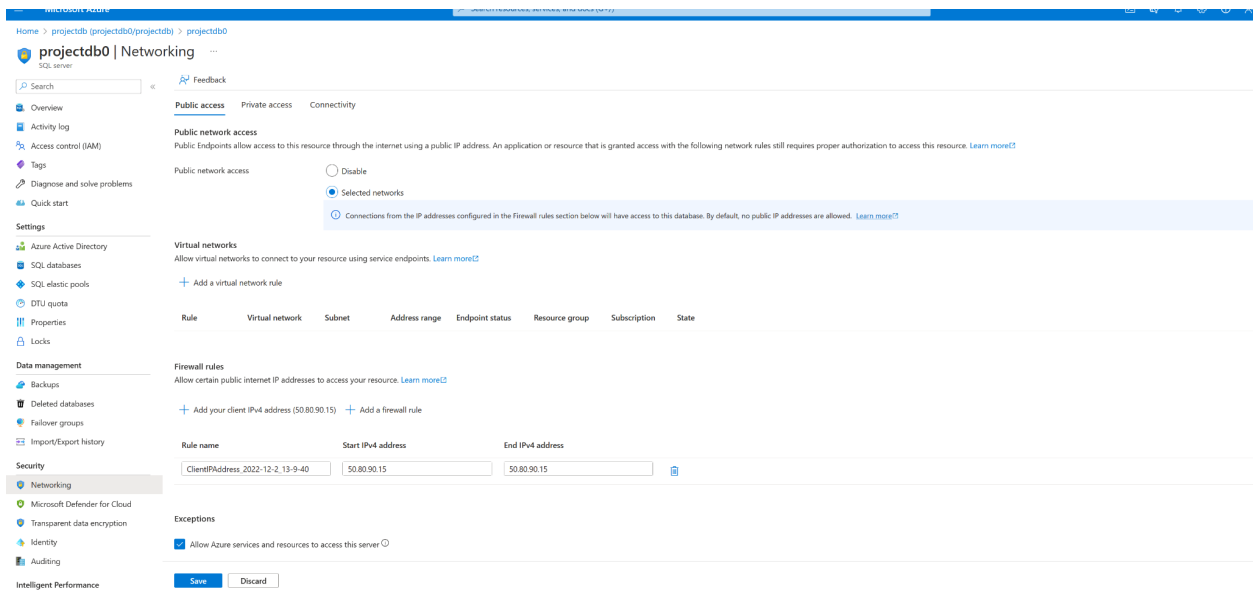
Then select a pipeline->select data flow.

From the dataflow, create a linked service for existing sql server instance.

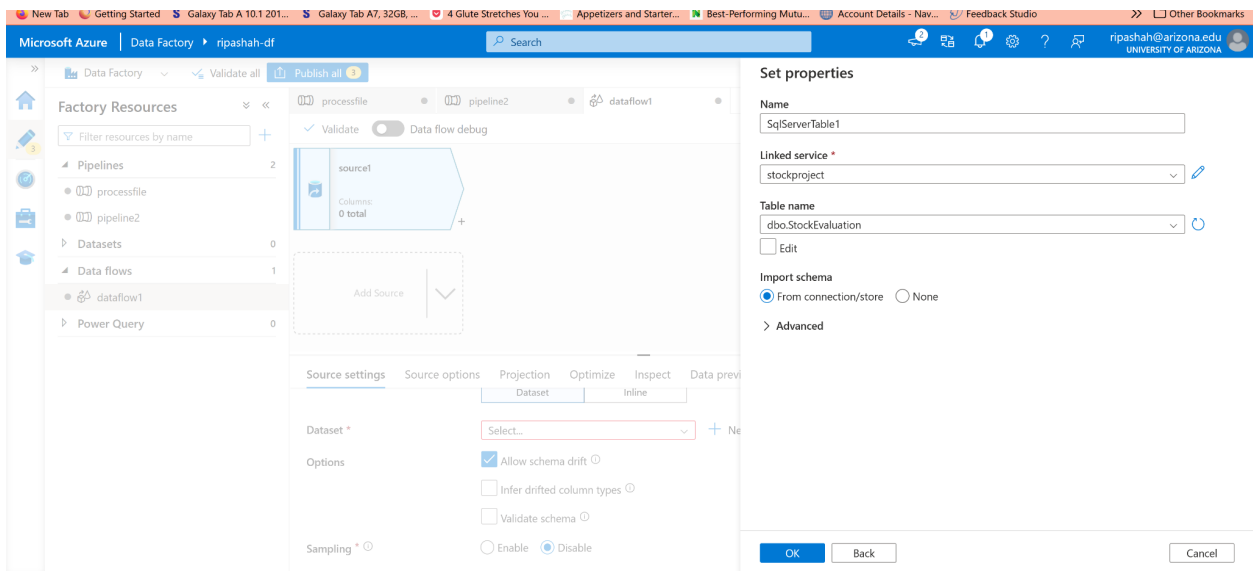


Check the firewall settings (network settings) for projectdb0 server:

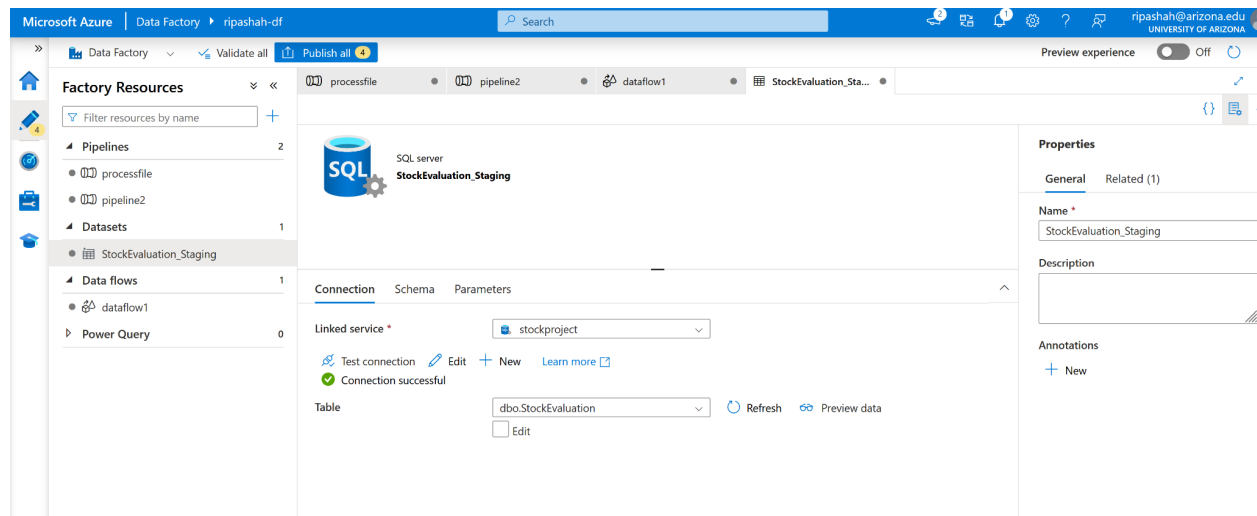




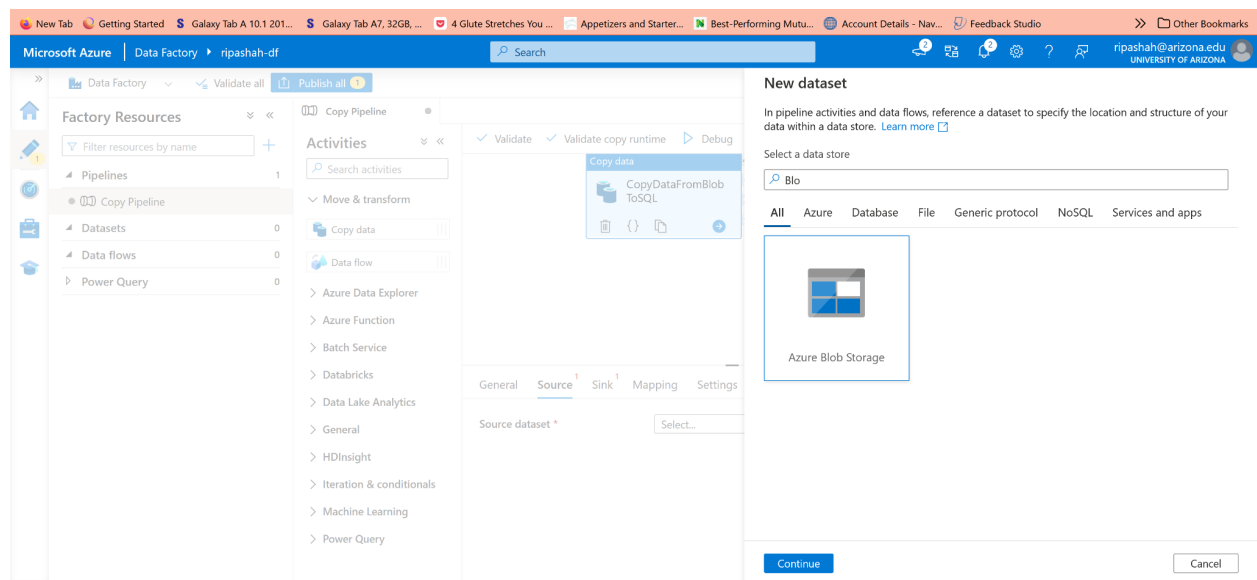
After changing the setting, please go back to the azure data factory. Please select the table from the list:

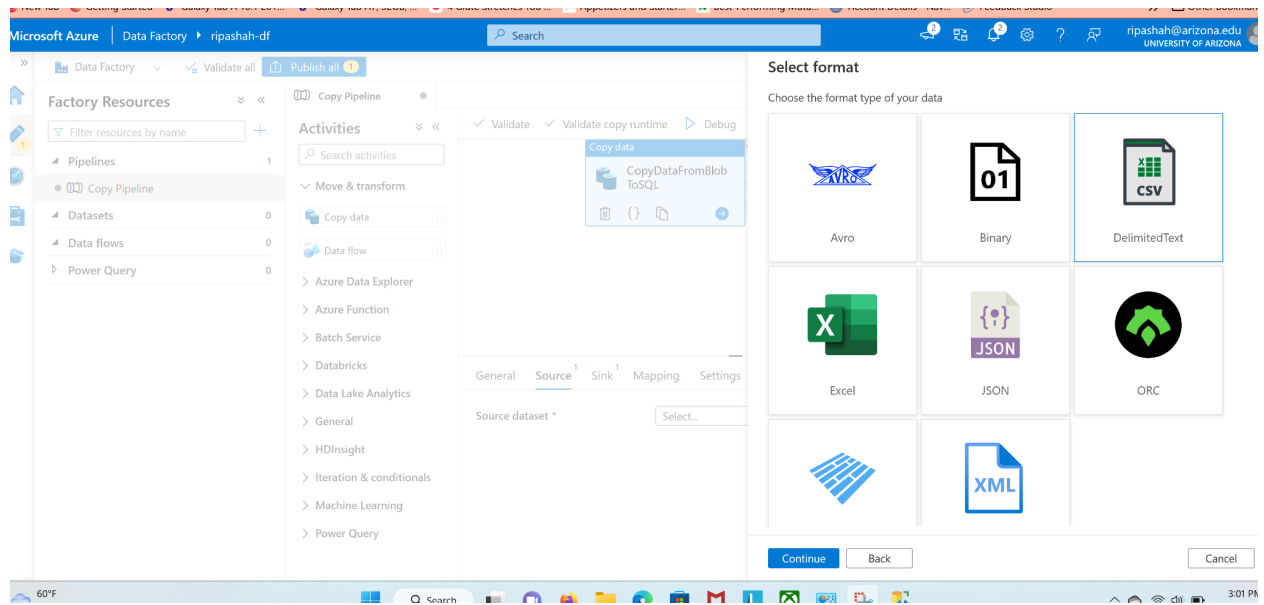


After selecting the table, you are able to check the connection.

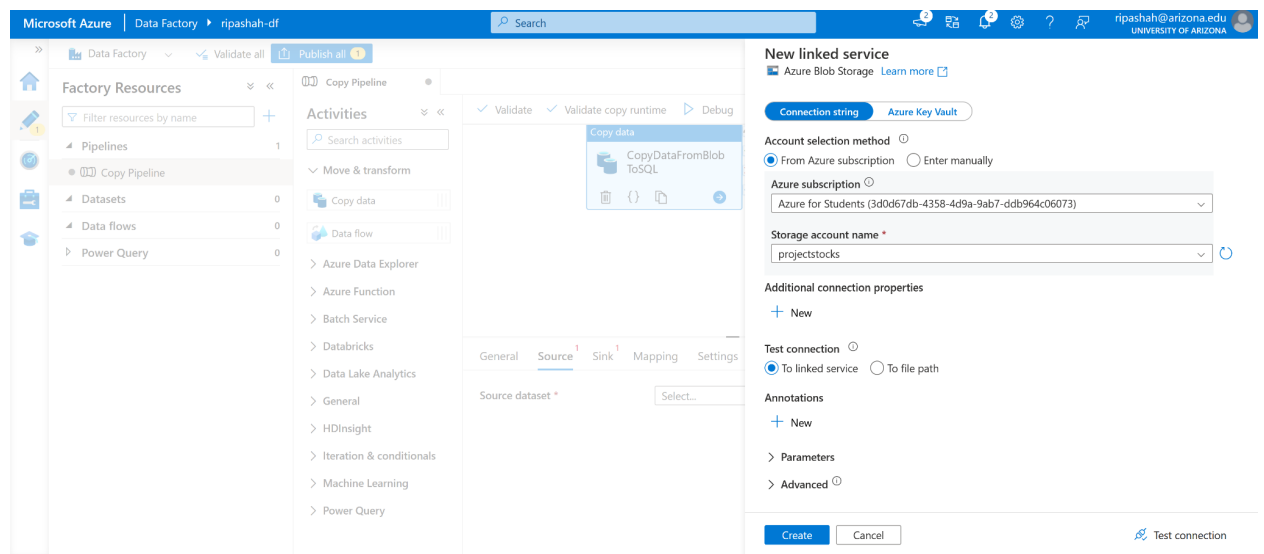


Now create a pipeline in the azure data factory. In that select activities->move and transform-> copy data. In the source section, select blob storage

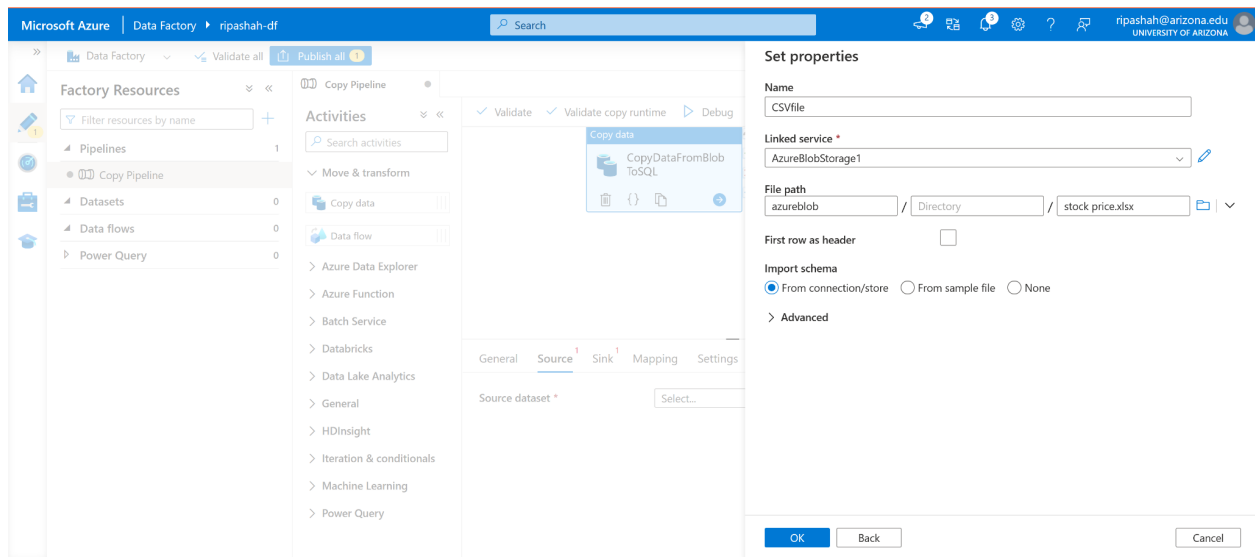
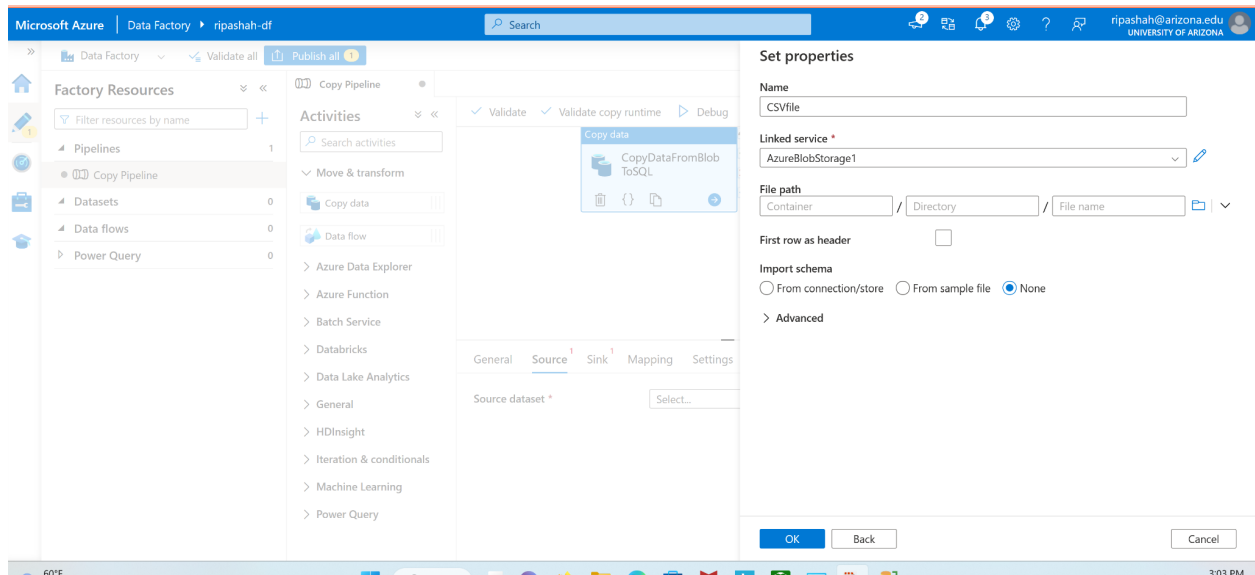




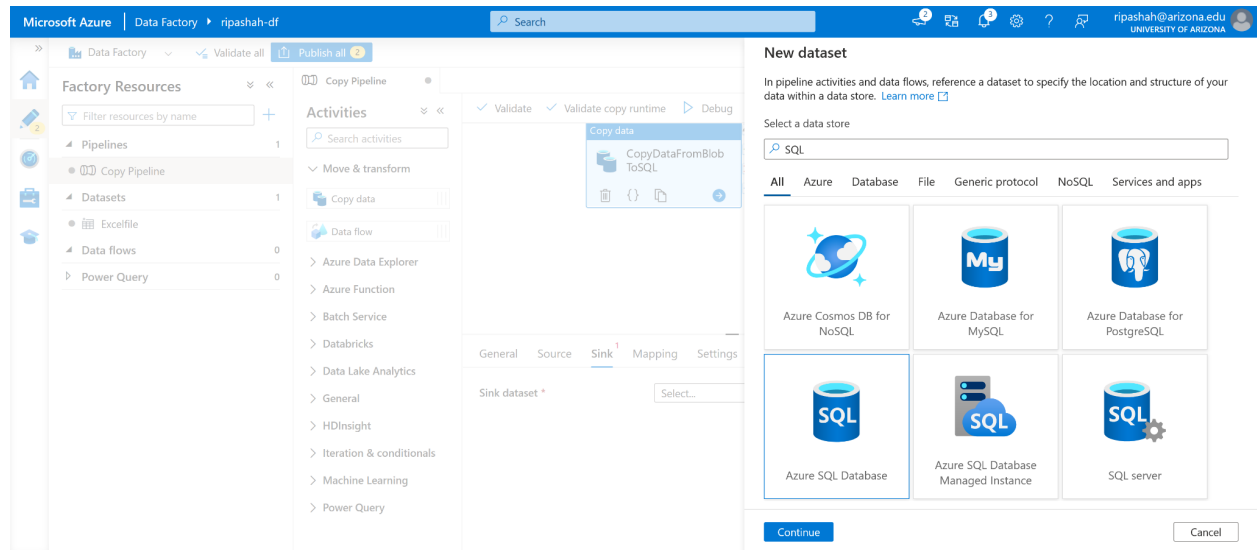
Now create a linked service to connect with the blob container



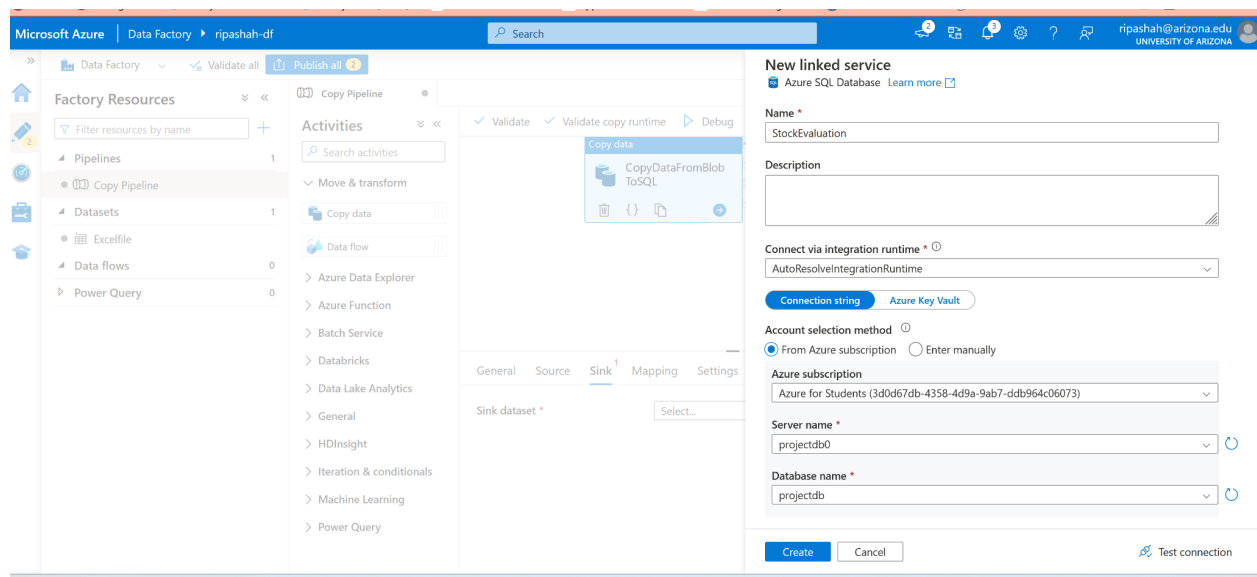
It will pop up set the properties window:

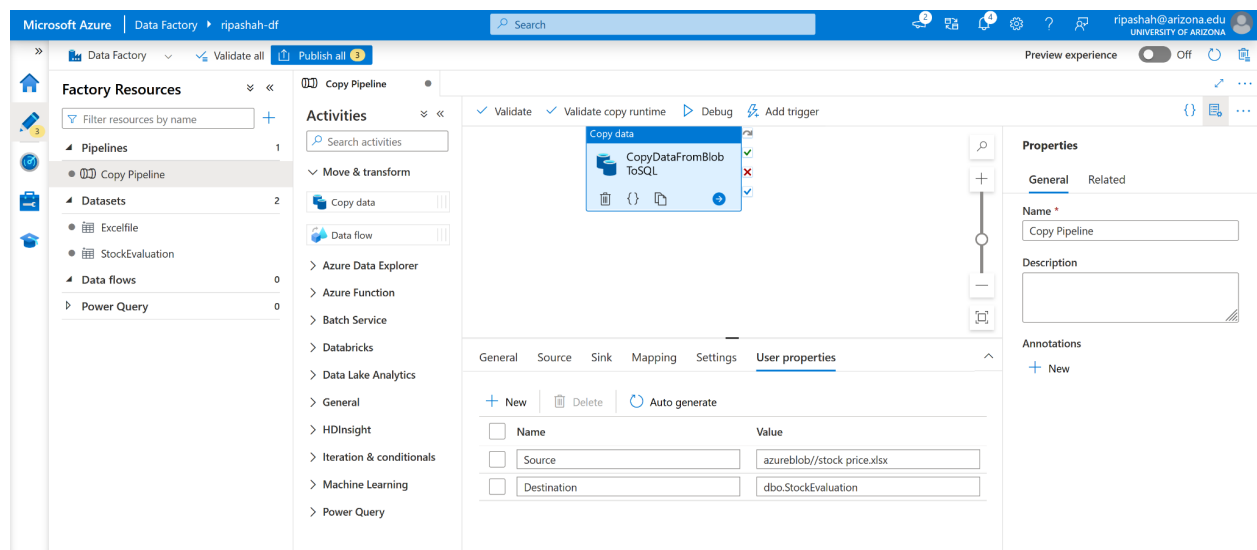
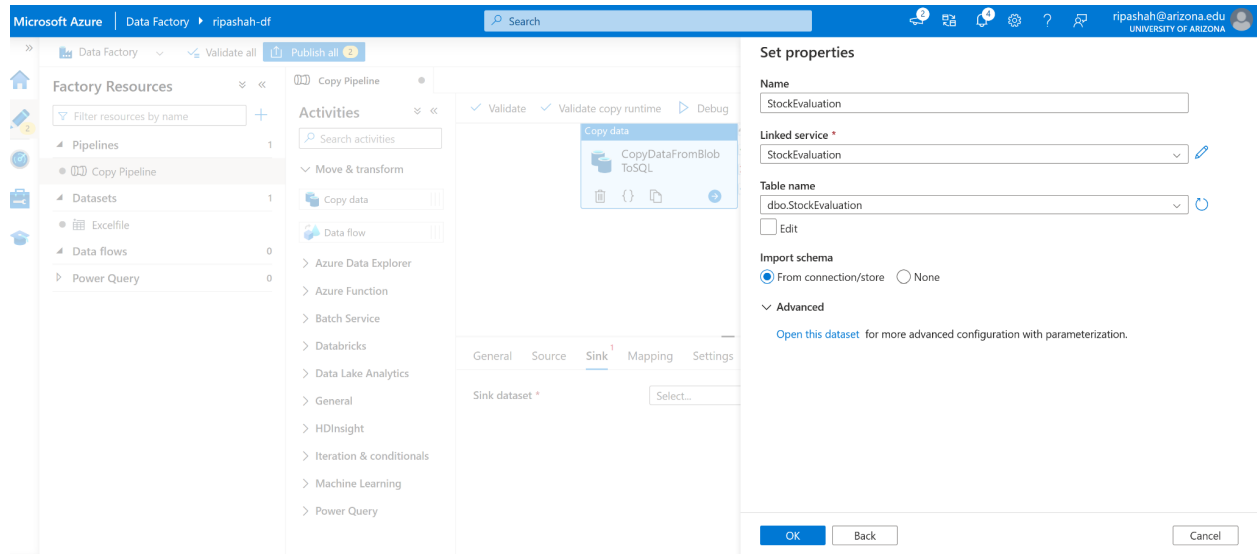


After completion, select the SQL database for the target

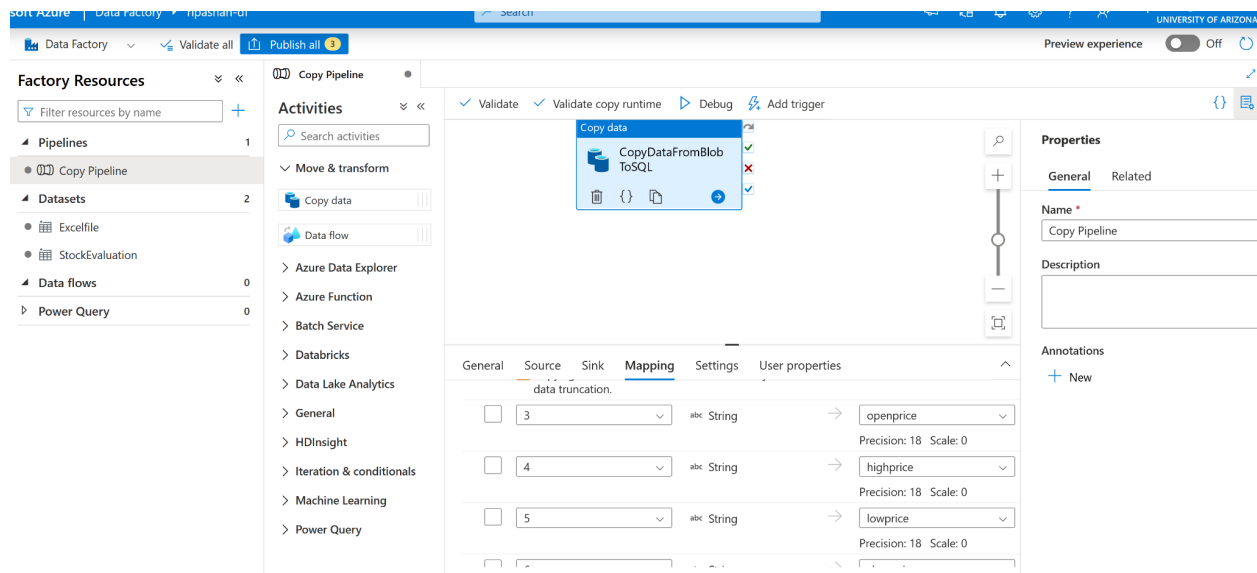


Create a linked service for the SQL database

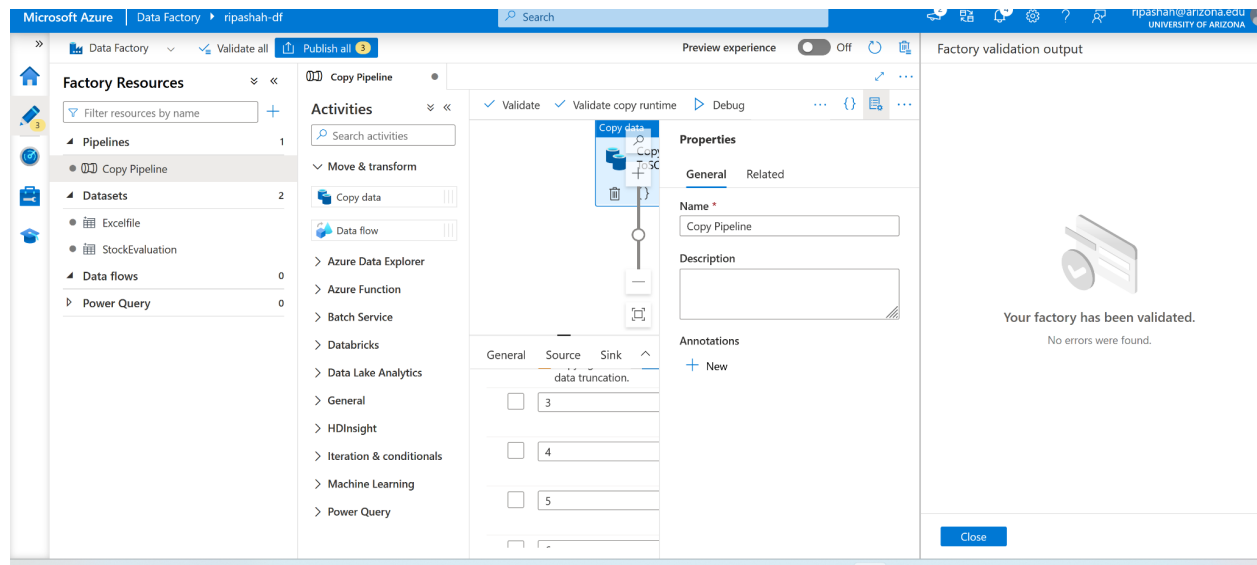




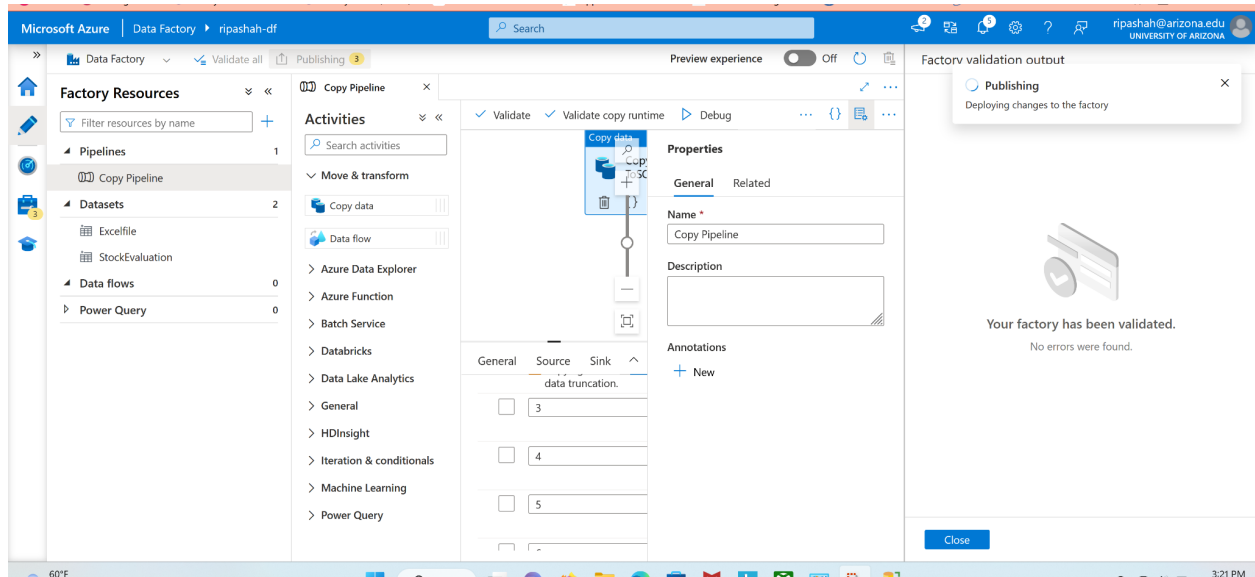
Check the mapping for both files:



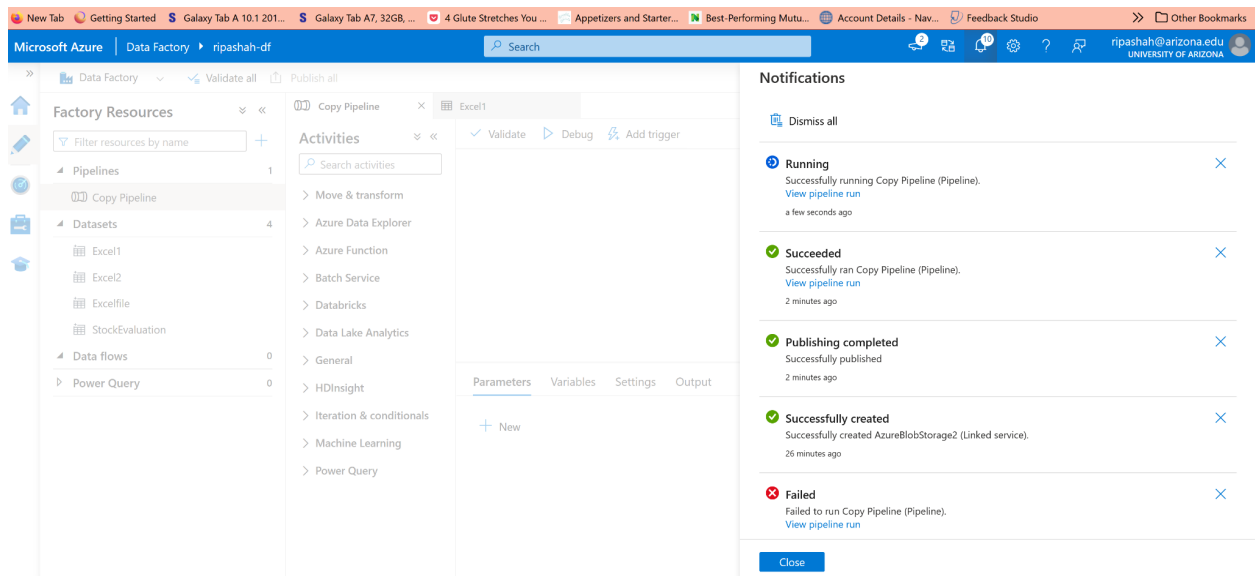
To validate, select the validate all from the top left corner



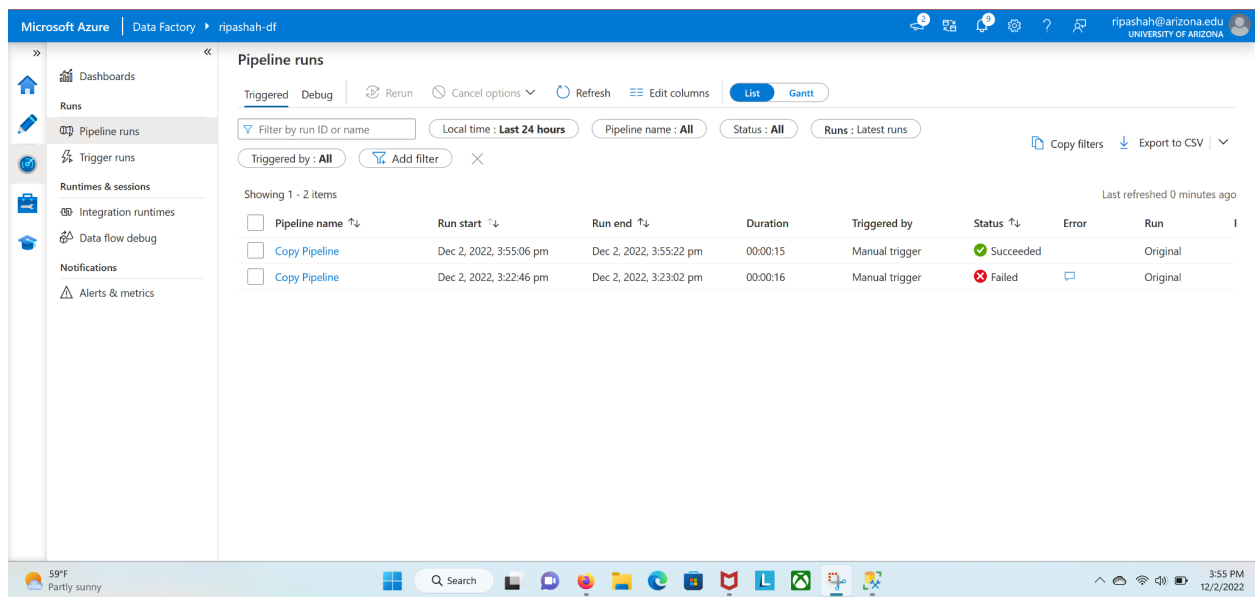
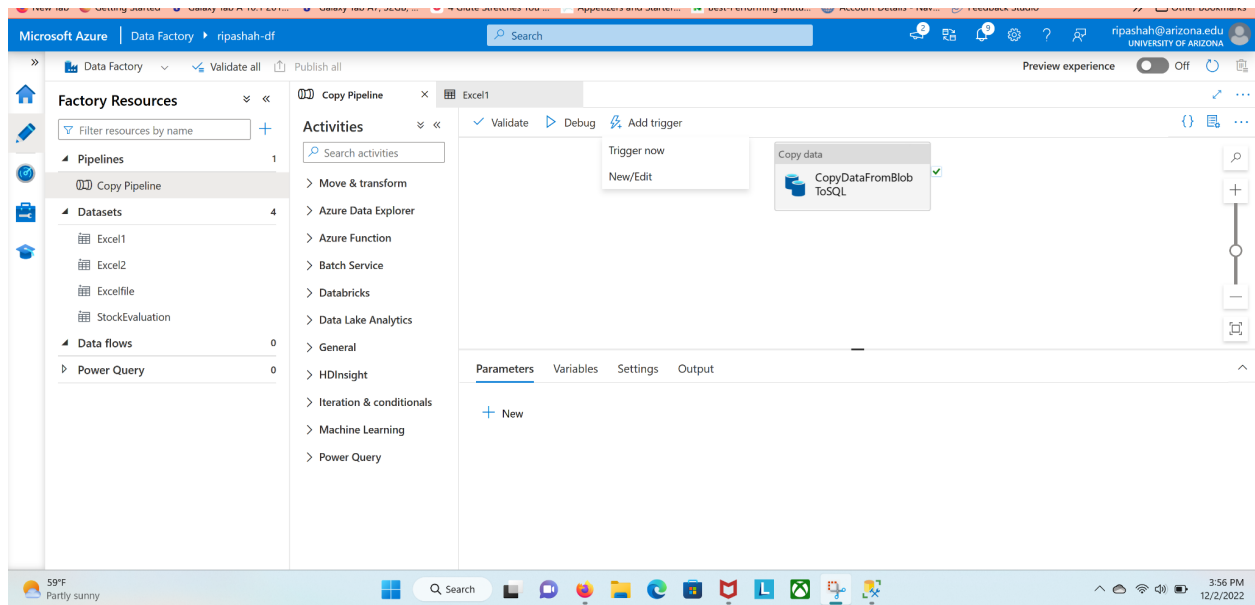
Once the pipeline runs successfully, in the top toolbar, select Publish all. Publishes entities (datasets, and pipelines) you created in Data Factory. Select Publish.



Select Trigger on the toolbar, and then select Trigger Now. On the Pipeline Run page, select OK.



Once it finishes publishing, you can add the trigger. You can check the status in monitor window



To see if the activity runs associated with the pipeline run, select the CopyPipeline link under the PIPELINE NAME column.

Select All pipeline runs at the top to go back to the Pipeline Runs view. To refresh the view, select Refresh.

Apply same step for the Amazon, TESLA, Facebook, Google and Netflix.

For single sheet, we should only use the above steps. However, we have multiple sheets so we will apply following logic:

Create a dataset for the excel file and set the dynamic parameters

all [Publish all](#)

Copy Pipeline pl_Multipleexcel MultipleExcelSheet StockEvaluation

Excel
MultipleExcelSheet

Connection Schema Parameters

Linked service * AzureBlobStorage1 [Test connection](#) [Edit](#) [+ New](#) [Learn more](#)

File path * azureblob / Directory / stock_price_id.xlsx [Browse](#)

Compression type None

Worksheet mode ☒ Name ☐ Index

Sheet name [ⓘ](#) @dataset().dsSheetName [Preview data](#)

Range [ⓘ](#) e.g. A3:H5

Null value

First row as header ☒

Copy Pipeline | pl_Multiplexcel | MultipleExcelSheet | StockEvaluation

Excel
MultipleExcelSheet

Connection Schema Parameters

Import schema Close

Column name	Type
id	String
ticker	String
Date	String
Open	String
High	String
Low	String
Close	String
Volume	String

Publish all | Copy Pipeline | pl_Multiplexcel | MultipleExcelSheet | StockEvaluation

Excel
MultipleExcelSheet

Connection Schema Parameters

+ New Delete

Name	Type	Default value
dsSheetName	String	Apple Stocks

Now create the dataset for the SQL table.

h-df

Search

Publish all

Copy Pipelinepl_MultipleExcelMultipleExcelSheetStockEvaluation

2

4

0

0

SQL

Azure SQL Database

StockEvaluation

Connection

Schema

Parameters

Linked service *

StockEvaluation

Test connection

Edit

New

Learn more

Connection successful

Table

dbo.StockEvaluation

Refresh

Preview data

Edit

Copy Pipelinepl_MultipleExcelMultipleExcelSheetStockEvaluation

SQL

Azure SQL Database

StockEvaluation

Connection

Schema

Parameters

Load schema

Clear

Column name	Type
ticker	varchar
stockid	date
openprice	decimal
highprice	decimal
lowprice	decimal
closeprice	decimal
volume	decimal
id	varchar

Now create a pipeline called pl_MultipleExcel.

publish all

Copy Pipelinepl_MultipleexcelMultipleExcelSheetStockEvaluation

SQL Azure SQL Database StockEvaluation

ConnectionSchemaParameters

Support schemaClose

Column name	Type	
ticker	varchar	
stockdt	date	
openprice	decimal	View details
highprice	decimal	View details
lowprice	decimal	View details
closeprice	decimal	View details
volume	decimal	View details
id	varchar	

publish all

Copy Pipelinepl_MultipleexcelMultipleExcelSheetStockEvaluation

Activities

store

General

Stored procedure

Lookup

Listofexcelshheet

ForEach

ForEach1

Activities

Copy data1

ParametersVariablesSettingsOutput

+ NewDelete

Name	Type	Default value
Variables	Array	[{"SheetName":"Apple Stocks"}]

Now create the variable called variable and put following options:
[{"SheetName":"Apple Stocks"}, {"SheetName":"Facebook Stocks"}],

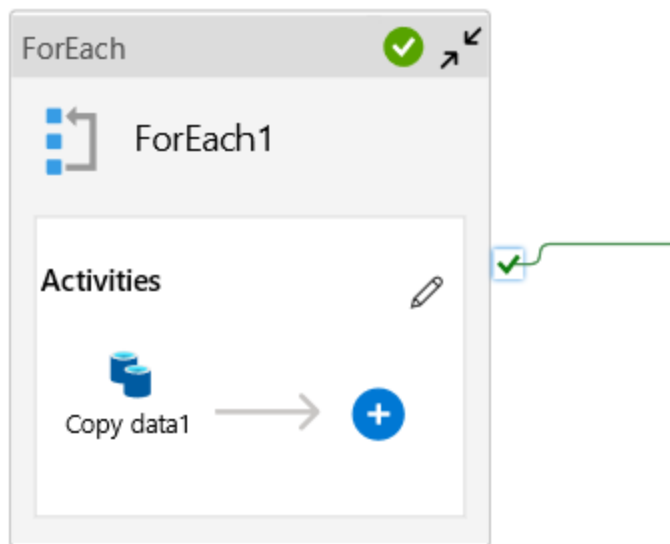
```
{"SheetName":"Netflix Stocks"},  
{"SheetName":"TESLA Stocks"}, {"SheetName":"Amazon Stocks"},  
{"SheetName":"Google Stocks"}]
```

Use variable for loop settings

The screenshot shows the 'Settings' tab of a workflow editor. It has four tabs: 'General', 'Settings' (selected), 'Activities (1)', and 'User properties'. Under the 'Settings' tab, there are three settings: 'Sequential' with an unchecked checkbox, 'Batch count' with a text input field and a help icon, and 'Items' with a dropdown menu showing '@variables(Variables)'.

Setting	Value
Sequential	☐
Batch count ⓘ	
Items	@variables(Variables)

Create the copy data activity for each loop.



In the source tab of copy data activity, use the multipleExcelSheet dataset as a source

Create dsSheetname parameter in the MultipleExcelSheet dataset

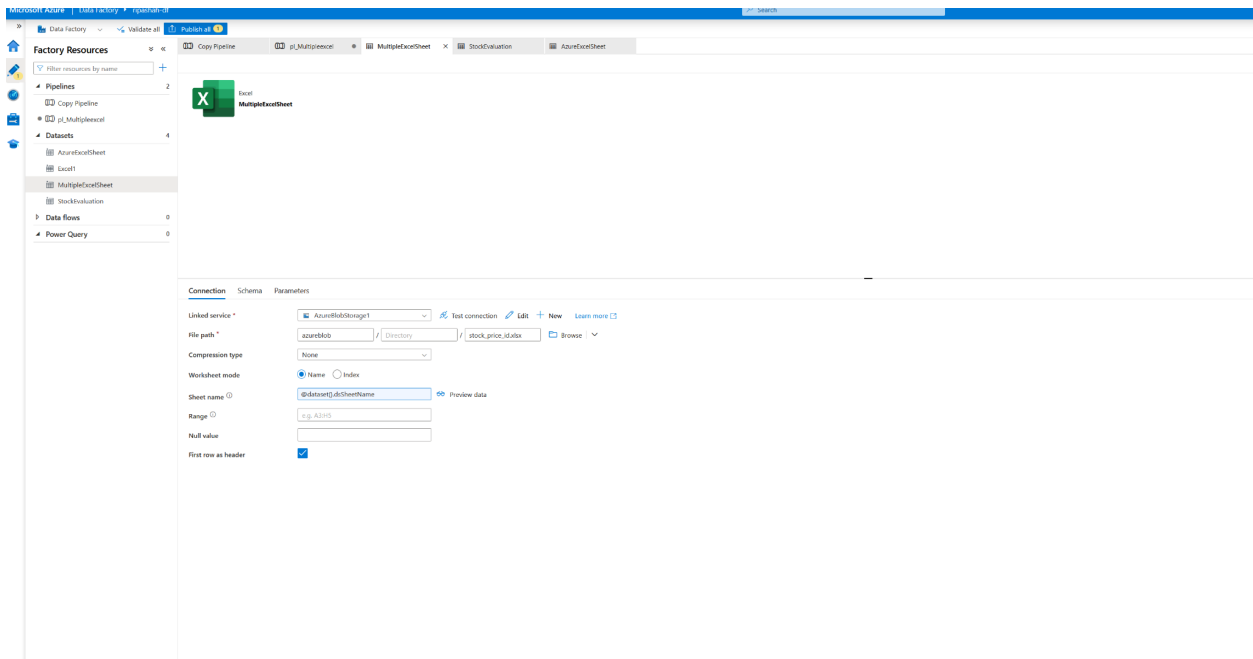
The screenshot shows the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane is visible, showing a tree view of resources including Pipelines, Datasets, Data flows, and Power Query. The 'MultipleExcelSheet' dataset is selected. The main pane displays the 'MultipleExcelSheet' dataset configuration. The 'Parameters' tab is active, showing a table with columns 'Name', 'Type', and 'Default value'. A parameter named 'dsSheetname' is listed with a 'String' type and a default value of 'Apple Stocks'.

Name	Type	Default value
dsSheetname	String	Apple Stocks

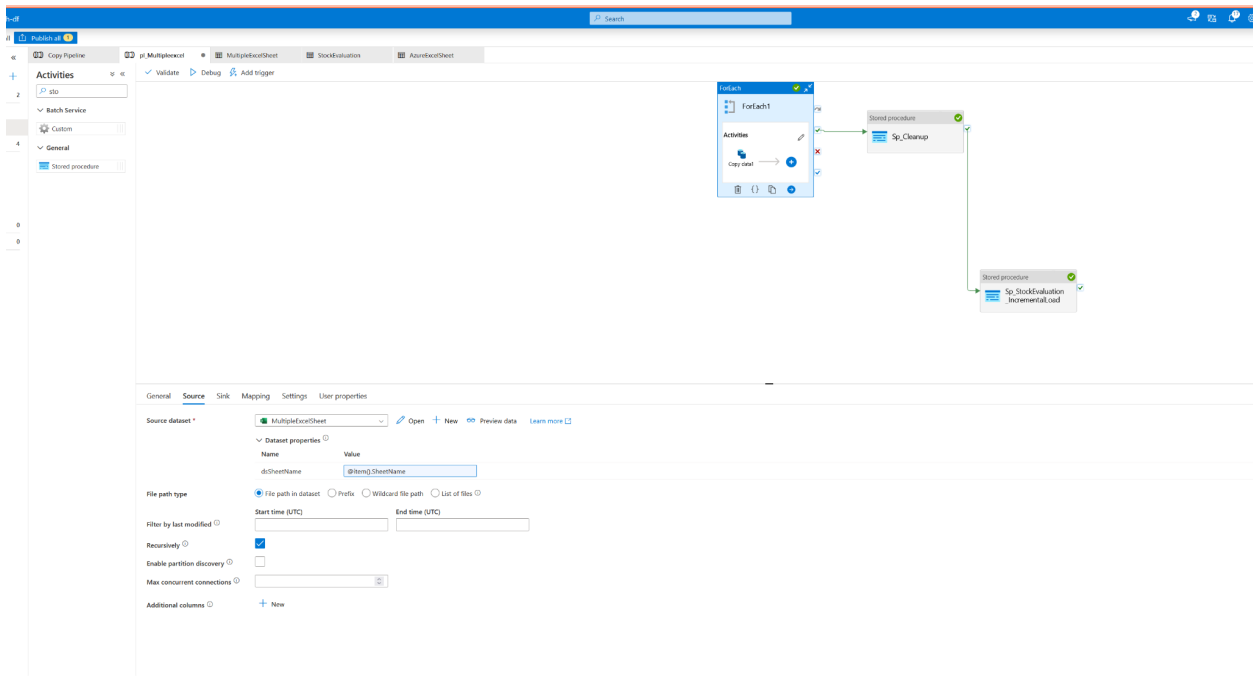
The screenshot shows the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane is visible, showing a tree view of resources including Pipelines, Datasets, Data flows, and Power Query. The 'MultipleExcelSheet' dataset is selected. The main pane displays the 'MultipleExcelSheet' dataset configuration. The 'Schema' tab is active, displaying a table with columns 'Column name' and 'Type'. The table lists various stock-related columns: id, ticker, Date, Open, High, Low, Close, and Volume, all of which are of type 'String'.

Column name	Type
id	String
ticker	String
Date	String
Open	String
High	String
Low	String
Close	String
Volume	String

Use @dataset.dsSheetName as a SheetName in the MultipleExcelSheet dataset



Now link that parameter with the foreach loop and use @item().SheetName as dsSheetName value in the source tab of the copy data activity inside the for each loop.



Now use StockEvaluation dataset for the sink

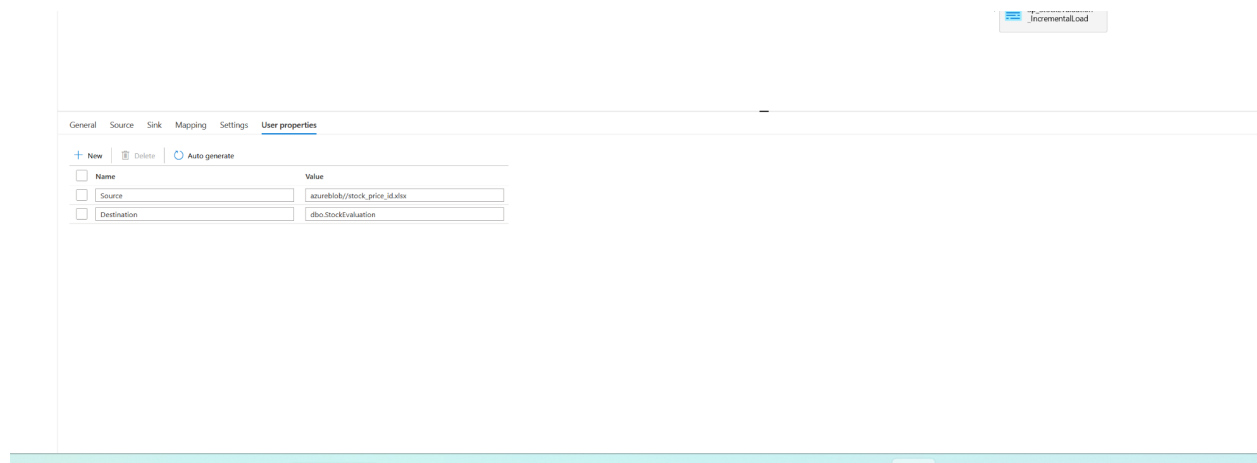
The screenshot shows the Azure Data Factory portal with the 'Sink' tab selected for the 'StockEvaluation' dataset. The 'Sink dataset' is set to 'StockEvaluation'. The 'Write behavior' is set to 'Insert'. The 'Bulk insert table lock' is set to 'No'. The 'Table option' is set to 'None'. The 'Pre-copy script' is empty. The 'Write batch timeout' is set to '30:00:00'. The 'Write batch size' is set to '100000'. The 'Max concurrent connections' is set to '10'. The 'Disable performance metrics analytics' checkbox is unchecked.

We can map source and sink as following:

The screenshot shows the Azure Data Factory portal with the 'Mapping' tab selected for the 'StockEvaluation' dataset. The 'Type conversion settings' are displayed, showing a table with source and destination columns and their types.

Source	Type	Destination	Type
Id	String	Id	varchar
Ticker	String	Ticker	varchar
Date	String	stockdt	date
Open	String	openprice	decimal
High	String	highprice	decimal
Low	String	lowprice	decimal
Close	String	closeprice	decimal

Please verify again source and destination using properties option



Once you successfully finish with the foreach activity, create the sp_cleanup stored procedure to remove null values from the StockEvaluation table

```

/***** Object:  StoredProcedure [dbo].[sp_cleanup]  Script Date: 12/4/2022
7:08:22 PM *****/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
/***** Script for SelectTopNRows command from SSMS *****/
ALTER procedure [dbo].[sp_cleanup]
as
begin
    select * FROM [dbo].[StockEvaluation]
    select * from [dbo].StockEvaluation_Target
    select * from dbo.StockEvaluation where id is null
    delete from StockEvaluation where id is null;
    delete from StockEvaluation where ticker is null;
End;

```

Once you execute sp_cleanup successfully, execute PrStockchurn for the incremental refresh technique.

```

/***** Object:  StoredProcedure [dbo].[PrStockChurn] Script Date: 12/4/2022
7:09:57 PM *****/
SET ANSI_NULLS ON
GO

```

```

SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[PrStockChurn]
as
BEGIN

--_*****
*****

--DECLARE @VarCurrentTimestamp as TIMESTAMP=CURRENT_TIMESTAMP;--this
built-in function might vary from database to database
--DECLARE @VarCurrentTimestamp as
TIMESTAMP=convert(timestamp,CURRENT_TIMESTAMP);
DECLARE @VarCurrentTimestamp as datetime2=convert(datetime2,getdate());
DECLARE @VarSourceRowCount as integer=0;
declare @VarTargetRowCount as integer=0;
declare @VarThresholdNbr as integer=0;
DECLARE @VarTinyIntVal as integer=0; --Note: TINYINT datatype can hold
values between 0 and 255 only. Any value assignment outside this range
would cause the stored procedure failure.
declare @tempdb as table(cnt int)
--_*****
*****

SELECT @VarSourceRowCount=COUNT(*)
FROM dbo.StockEvaluation; --Replace this with actual database name and
table name (e.g., CustomerChurn_Stage) that you use.
select @VarSourceRowCount;
SELECT @VarTargetRowCount=count(*)
FROM dbo.StockEvaluation_Target; --Replace this with actual database name
and table name (e.g., CustomerChurn) that you use.
select @VarTargetRowCount
-- (TargetCount * 20%)
SELECT
@VarThresholdNbr=CAST((@VarTargetRowCount * .2) AS INTEGER)
FROM @tempdb; -- The DUMMY is system table which might vary from database
to database. For your database, you need to figure out.
select @VarThresholdNbr
--_*****
*****

--Fail the Stored Proc if the Source Row Count is less than the Threshold
Number (i.e., 20% of the Target Table row count).
--This ensures that the Target table is not refreshed with incomplete set

```

of Source Data

```
IF (@VarSourceRowCount < @VarThresholdNbr )

SELECT @VarTinyIntVal=-1
FROM @tempdb;
select @VarTinyIntVal
--_*****
*****
--_*****
*****

-- DELETE target table rows which are no longer available in source
database table.

DELETE
FROM dbo.StockEvaluation_Target
WHERE EXISTS
(

SELECT *
FROM
(
SELECT dbo.StockEvaluation_Target.id FROM dbo.StockEvaluation_Target --AS
TrgtTbl -- Example table name: CustomerChurn

EXCEPT

SELECT dbo.StockEvaluation.id FROM dbo.StockEvaluation --AS SrcTbl
--Example table name: CustomerChurn_Stage

) AS SrcTbl

WHERE StockEvaluation_Target.id = SrcTbl.id)
--AND TrgtTbl.Primary Key Column2 = SrcTbl.Primary Key Column2
--AND ...
--_*****UPDATE ROWS THAT CHANGED IN
SOURCE*****
-- Update the rows for which new version of rows have arrived as part of
delta/incremental feed (i.e., change to non-key values).
update
dbo.StockEvaluation_Target
SET StockEvaluation_Target.id = StockEvaluation.id,
    StockEvaluation_Target.ticker= StockEvaluation.ticker,
```

```

        StockEvaluation_Target.stockdt = StockEvaluation.stockdt,
        StockEvaluation_Target.openprice=StockEvaluation.openprice,
        StockEvaluation_Target.highprice=StockEvaluation.highprice,
        StockEvaluation_Target.lowprice=StockEvaluation.lowprice,
        StockEvaluation_Target.closeprice=StockEvaluation.closeprice,
        StockEvaluation_Target.volume=StockEvaluation.volume,
        StockEvaluation_Target.SourceSystemNM=
convert(nvarchar(50),SERVERPROPERTY('MachineName')),
        StockEvaluation_Target.CreateDtm=@VarCurrentTimestamp,
        StockEvaluation_Target.CreateAgentId='test',
        StockEvaluation_Target.ChangeDtm = convert(datetime,GETDATE())
from dbo.StockEvaluation_Target
INNER JOIN
dbo.StockEvaluation

```

```

on  StockEvaluation_Target.id = StockEvaluation.id

```

```

WHERE (
        COALESCE(StockEvaluation_target.ticker, '*') <>
COALESCE(StockEvaluation.ticker, '*')
        OR COALESCE(StockEvaluation_target.stockdt, '*') <>
COALESCE(StockEvaluation.stockdt, '*')
        OR COALESCE(StockEvaluation_target.openprice, '*') <>
COALESCE(StockEvaluation.openprice, '*')
        OR COALESCE(StockEvaluation_target.highprice, '*') <>
COALESCE(StockEvaluation.highprice, '*')
        OR COALESCE(StockEvaluation_target.lowprice, '*') <>
COALESCE(StockEvaluation.lowprice, '*')
        OR COALESCE(StockEvaluation_target.closeprice, '*') <>
COALESCE(StockEvaluation.closeprice, '*')
        OR COALESCE(StockEvaluation_target.volume, '*') <>
COALESCE(StockEvaluation.volume, '*')
)

```

```

-----create CustomerChurn table-----

```

```

--CREATE TABLE [dbo].[CustomerChurn](
--    [CustomerId] [varchar](50) NOT NULL,
--    [Surname] [varchar](50) NULL,
--    [CreditScore] [int] NULL,
--    [Geography1] [nvarchar](50) NULL,
--    [Gender] [nvarchar](20) NULL,
--    [Age] [int] NULL,

```

```

--      [balance] [float] NULL,
--      [Exited] [nvarchar](50) NULL,
--      [SourceSystemNM] [nvarchar](50) NOT NULL,
--      [CreateAgentId] [nvarchar](20) NOT NULL,
--      [CreateDtm] [timestamp] NOT NULL,
--      [ChangeAgentId] [nvarchar](20) NOT NULL,
--      [ChangeDtm] [datetime] NOT NULL,
-- CONSTRAINT [PK_CustomerChurn_Stage] PRIMARY KEY CLUSTERED
--(
--      [CustomerId] ASC
--)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
--) ON [PRIMARY]
insert into [StockEvaluation_Target](
    [id],
    [ticker],
    [stockdt],
    [openprice],
    [highprice],
    [lowprice],
    [closeprice],
    [volume],
    [SourceSystemNM],
    [CreateAgentId] ,
    [CreateDtm] ,
    [ChangeAgentId],
    [ChangeDtm])
SELECT
StockEvaluation.id
,StockEvaluation.ticker
---,SrcTbl.Primary Key Column3 ... list all PK columns based on your
source/taget table primary key(s)
,StockEvaluation.stockdt
,StockEvaluation.openprice
,StockEvaluation.highprice
,StockEvaluation.lowprice
,StockEvaluation.closeprice
,StockEvaluation.volume
,convert(nvarchar(50),HOST_NAME()) AS "SourceSystemNm" --Non-Primary Key
,USER AS "CreateAgentId" --Non-Primary Key -----use you
own USERID/USERNAME depending on the database system you use
,convert(datetime2,@VarCurrentTimestamp) AS "CreateDtm" --Non-Primary Key

```

```

,USER AS "ChangeAgentId" --Non-Primary Key -----use you
own USERID/USERNAME depending on the database system you use
,convert(datetime2,@VarCurrentTimestamp) AS "ChangeDtm" --Non-Primary Key

FROM StockEvaluation

INNER JOIN
(
SELECT StockEvaluation.id

FROM StockEvaluation

EXCEPT

SELECT StockEvaluation_Target.id

FROM StockEvaluation_Target

) AS ChgdNew

ON StockEvaluation.id = ChgdNew.id
--AND SrcTbl.Primary Key Column2 = ChgdNew.Primary Key Column2
--AND ...

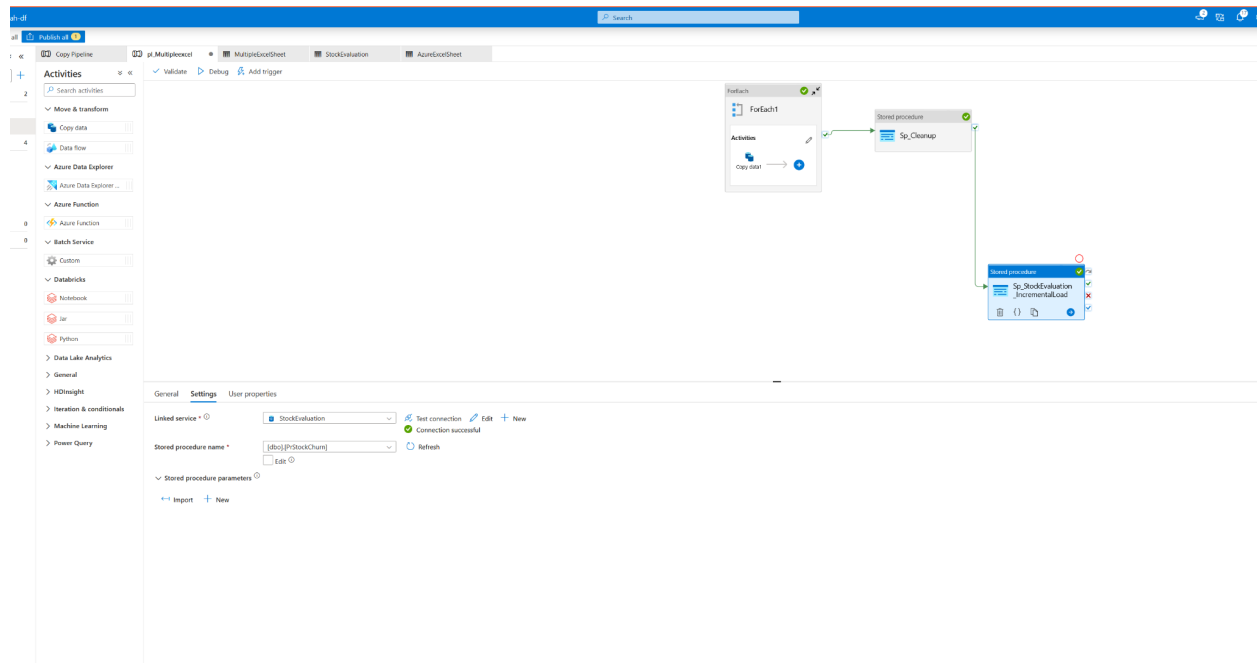
end;

print @VarSourceRowCount;
print @VarTargetRowCount;
print @VarThresholdNbr;
--select * from tempdb;

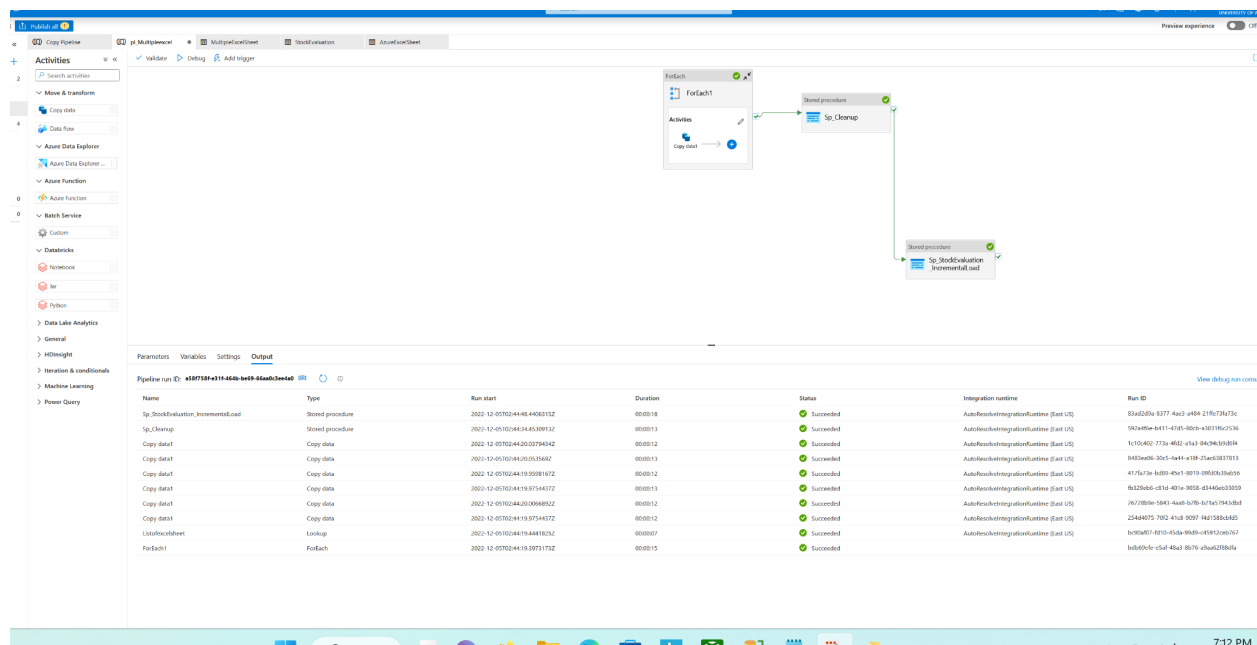
```

Now find the stored procedure activity in the pipeline (author tab)

Use StockEvaluation as the linked service. Also, select the dbo.PrStockChurn as the stored procedure name



Connect all the activities and publish them all according to the following screenshot



It will execute pl_MultipleExcel pipeline.

Name	Type	Run start	Duration	Status	Integration runtime	Run ID
Incremental refresh	Stored procedure	2022-12-03T04:46:45.445Z	00:00:15	Succeeded	AutoResolveIntegrationRuntime (Bat US)	85a3c3b6-6377-4a47-a834-714a75a75e7e
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:13	Succeeded	AutoResolveIntegrationRuntime (Bat US)	252a7f6e-9171-4710-8930-40217852233e
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:12	Succeeded	AutoResolveIntegrationRuntime (Bat US)	1c194d02-773a-49d7-41d3-84c14505004a
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:13	Succeeded	AutoResolveIntegrationRuntime (Bat US)	8d03a0b0-3635-4a48-a788-75a433078613
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:12	Succeeded	AutoResolveIntegrationRuntime (Bat US)	41797a-8889-47e1-8d79-0958b17d8d5e
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:13	Succeeded	AutoResolveIntegrationRuntime (Bat US)	8d17d0a0-07d4-47e1-8d79-0958b17d8d5e
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:13	Succeeded	AutoResolveIntegrationRuntime (Bat US)	207238ba-79d3-4a48-8786-527479433d8d
Copy Data	Copy data	2022-12-03T04:46:45.445Z	00:00:12	Succeeded	AutoResolveIntegrationRuntime (Bat US)	22a99075-1985-47e1-8d79-0958b17d8d5e
Lookup	Lookup	2022-12-03T04:46:45.445Z	00:00:07	Succeeded	AutoResolveIntegrationRuntime (Bat US)	3a70a077-1e1d-47e1-8d79-0958b17d8d5e
ForEach	ForEach	2022-12-03T04:46:45.445Z	00:00:15	Succeeded	AutoResolveIntegrationRuntime (Bat US)	3d8d8b1e-7d4d-4b79-8d79-0958b17d8d5e

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Error	Run	Parameters	Annotations	Run ID
pl_MultiRefresh	Dec 4, 2022, 6:13:19 pm	Dec 4, 2022, 6:14:34 pm	00:01:15	Manual trigger	Succeeded		Original			6700e089-6d0b-4b7a-a834-071096c8f5a
pl_MultiRefresh	Dec 4, 2022, 5:46:24 pm	Dec 4, 2022, 5:46:44 pm	00:00:19	Manual trigger	Succeeded		Original			2a2e7187-1f65-4232-8c91-88a09718b0a6
pl_MultiRefresh	Dec 4, 2022, 4:55:38 pm	Dec 4, 2022, 4:55:46 pm	00:00:08	Manual trigger	Failed		Original			8690739b-7e65-4445-8856-f8a0475072ae
pl_MultiRefresh	Dec 4, 2022, 4:50:09 pm	Dec 4, 2022, 4:50:17 pm	00:00:08	Manual trigger	Failed		Original			1779d4c2-9d0d-48e1-8a0a-ba739a7e00d0
Copy Pipeline	Dec 4, 2022, 1:13:26 pm	Dec 4, 2022, 1:13:36 pm	00:00:09	Manual trigger	Failed					
Copy Pipeline	Dec 4, 2022, 1:04:58 pm	Dec 4, 2022, 1:05:08 pm	00:00:09	Manual trigger	Failed					

From this you can apply incremental refresh in the azure data factory. You can also check the code in the following git repository <https://github.com/Ripa-Shah/Azure-Data-Factory/tree/secondary>

Here is the output:

SQLQuery19.sql - pr...objectdb (ssaa (71))* X SQLQuery18.sql - pr...objectdb (ssaa (80)) SQLQuery17.sql - pr...objectdb (ssaa (91))

```

select * from StockEvaluation;
select * from StockEvaluation_Target

```

100 %

Results Messages

	ticker	stockdt	openprice	highprice	lowprice	closeprice	volume	id	
270437	TESLA	2022-11-22	168	170	166	169	78452327	TESLA16198	
270438	TESLA	2022-11-23	173	183	172	183	109536...	TESLA16199	
270439	TESLA	2022-11-25	185	185	180	182	50672739	TESLA16200	
270440	TESLA	2022-11-28	179	188	179	182	93038148	TESLA16201	
270441	TESLA	2022-11-29	184	186	178	180	83357111	TESLA16202	
270442	TESLA	2022-11-30	182	194	180	194	109186...	TESLA16203	
270443	TESLA	2022-12-01	197	198	191	194	80046213	TESLA16204	
270444	TESLA	2022-12-02	191	196	191	194	73645922	TESLA16205	

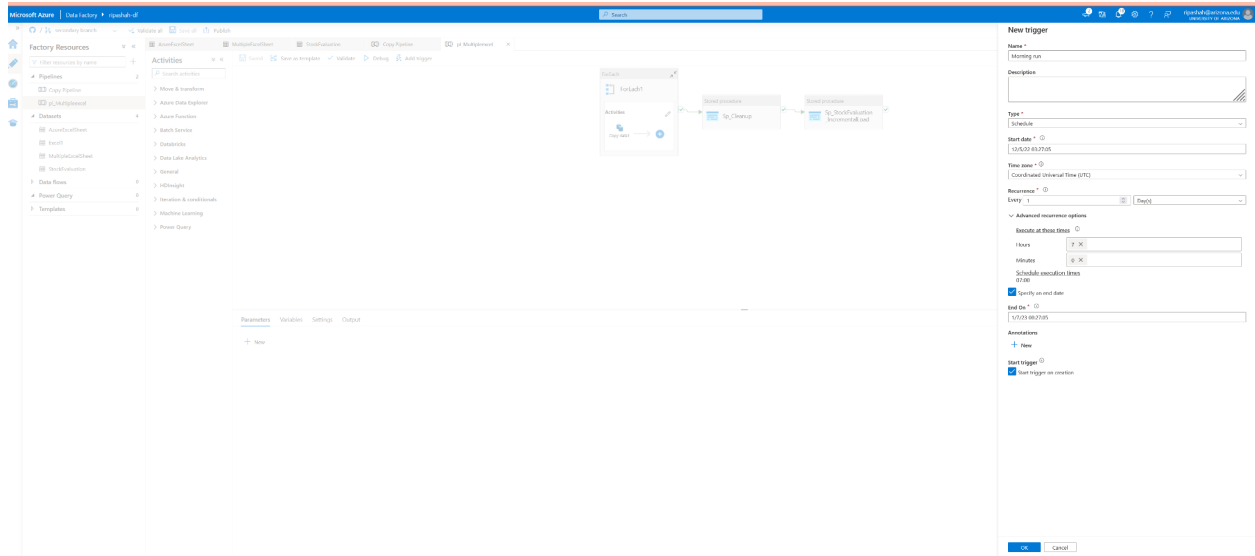
	ticker	stockdt	openprice	highprice	lowprice	closeprice	volume	SourceSystemNM	CreateAgentId	CreateDtm	Ch
270437	TESLA	2022-11-22	168	170	166	169	78452327	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270438	TESLA	2022-11-23	173	183	172	183	109536...	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270439	TESLA	2022-11-25	185	185	180	182	50672739	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270440	TESLA	2022-11-28	179	188	179	182	93038148	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270441	TESLA	2022-11-29	184	186	178	180	83357111	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270442	TESLA	2022-11-30	182	194	180	194	109186...	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270443	TESLA	2022-12-01	197	198	191	194	80046213	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc
270444	TESLA	2022-12-02	191	196	191	194	73645922	aaef2317200000G	dbo	2022-12-05 02:44:50.9766667	dbc

Query executed successfully. projectdb0.database.windows... ssaa (71) projectdb 00:00:05 270,444 rows

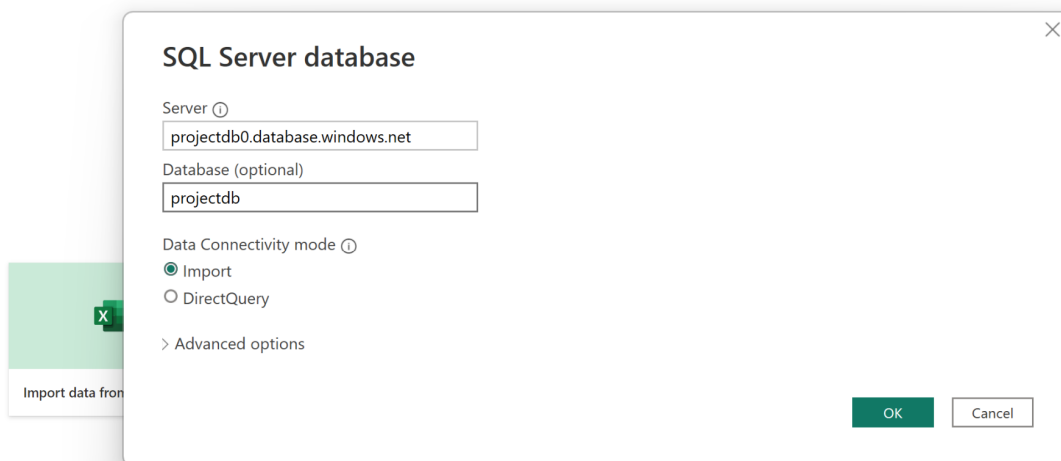
StockEvaluation_Target table has been updated successfully. Also, Source systemNM, CreateAgentId, CreateDtm, ChangeAgentID and ChangeDtm have been updated successfully.

According to the excel file, it gives everyday updates of the stock market. I have used Facebook, Amazon, Apple, Netflix, Google (FAANG) and Tesla stocks in the excel file

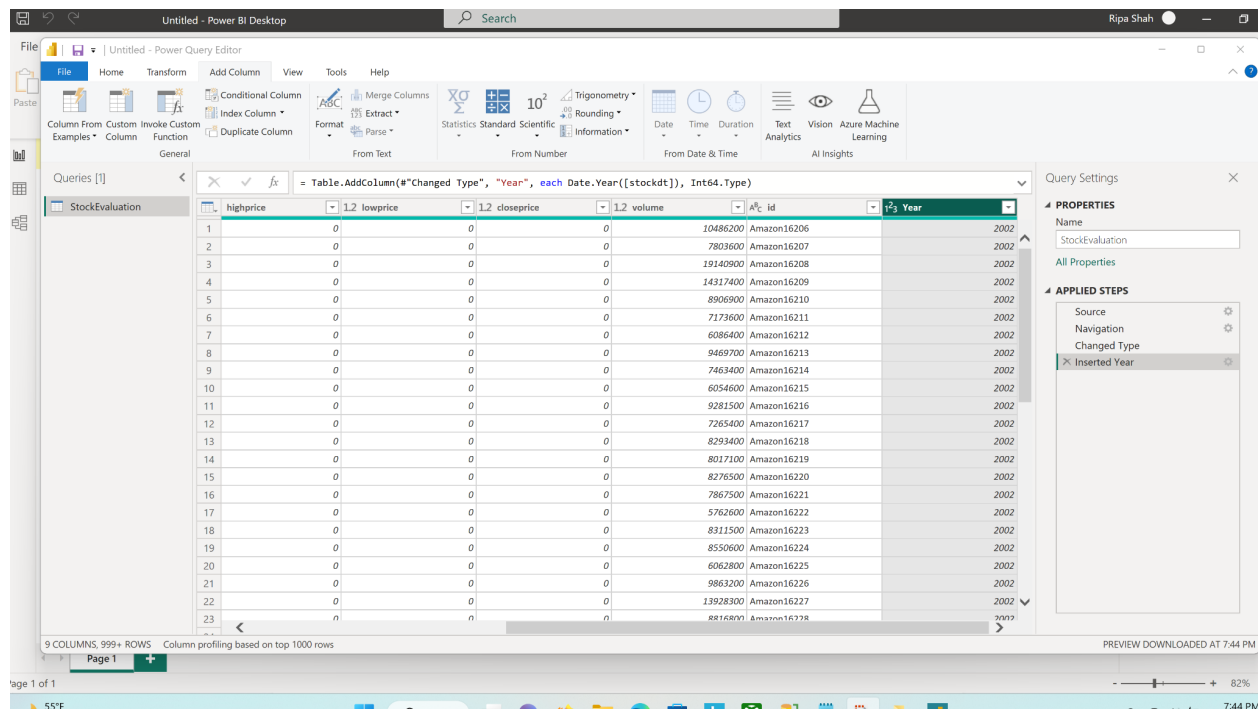
It will update everyday at 7:00 am using trigger:



For the power bi project, I will use the following connections: projectdb0.database.windows.net



After connecting with the Azure SQL database, I'm using StockEvaluation table to transform/clean the data.



Once you finish updating the power query, select the close and apply from the file menu option

In the following screenshot, I have created different graph to analyze the previous year, previous three years data. In that, I have created two slicer (1) for stocks (2) year. This way anyone could analyze the performance of stocks of previous year.



In the text file, I am attaching a code for tables and stored procedures. Also, I am attaching the power bi file. Also, please check github for more information.

<https://github.com/Ripa-Shah/PowerBI>

<https://github.com/Ripa-Shah/Azure-Data-Factory>